IntentCloud: Intent-Aware Cognitive Cloud Memory System

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***Abstract*—Conventional file storage systems, including Google Drive, Dropbox, and local hierarchical file systems, organize documents primarily through filenames and folder structures. As the volume of stored files increases, locating a specific document becomes progressively more difficult because users often cannot recall exact filenames, storage locations, or the precise keywords used at the time of creation. To address this limitation, this paper proposes IntentCloud, an Intent-Aware Cognitive Cloud Memory system that enables semantic document retrieval through natural language queries. Rather than relying on filename- or keyword-based search, users can describe the content, purpose, or context of a document, allowing the system to identify relevant files based on semantic meaning rather than lexical overlap. IntentCloud combines a locally hosted small language model for intent understanding, transformer-based sentence embeddings for semantic representation, and a vector similarity index for efficient retrieval. Persistent storage is maintained on a dedicated edge node, while a secure tunneling mechanism enables remote access without exposing the underlying network. The system supports cross-platform hardware acceleration, ensuring compatibility across a range of consumer-grade computing devices. Unlike cloud-dependent Retrieval-Augmented Generation (RAG) solutions, IntentCloud performs text extraction, embedding generation, and intent analysis entirely on local hardware, thereby improving privacy, minimizing dependence on third-party services, and enabling secure, intent-aware document retrieval. The proposed architecture is evaluated against nineteen recent semantic retrieval and RAG-based approaches drawn from the literature.**

***Index Terms*—semantic search, cognitive memory, natural language retrieval, vector similarity search, large language models, edge computing, privacy-preserving retrieval, cross-platform hardware acceleration, Retrieval-Augmented Generation**

I. Introduction

Contemporary file storage systems in widespread use today, such as Google Drive, Dropbox, Notion, and local hierarchical folder structures, primarily rely on filename-based indexing and manual folder organization. This design implicitly assumes that a user will accurately recall, months or years later, the precise name assigned to a file or the specific folder in which it was placed. In practice, this assumption does not hold: as the number of stored files grows, users become progressively less able to locate documents by name, spend disproportionate time navigating nested folder hierarchies, and encounter the well-documented limitations of keyword-based search. Keyword matching frequently returns an excess of irrelevant results or fails entirely to surface semantically related documents whenever the search term does not appear verbatim in the target file [1].

Consider a scenario in which a student has accumulated more than two hundred files over the course of a semester. Several months later, the student is required to produce a report concerning Kafka and distributed systems but can no longer recall the filename, the date of upload, or the folder in which the document was stored. A literal keyword search for "Kafka" returns dozens of unrelated results, none of which correspond to the desired file. A semantically expressed query, such as "the report where I discussed Kafka and microservices," should instead surface the correct document directly. This retrieval gap between what a user remembers and how conventional systems index information constitutes the central problem that IntentCloud is designed to address.

IntentCloud is a self-hostable, artificial-intelligence-enhanced file storage and retrieval system that provides intelligent and privacy-preserving document management. When a file is uploaded, the system first extracts its textual content and subsequently transforms this text into semantic embeddings, which are stored in a vector index. When a user submits a query expressed in natural language, the system infers the underlying intent of the query and performs semantic similarity matching against the stored embeddings to identify the most relevant files. The requested document is then retrieved from persistent edge storage and made available for download. To achieve scalability and computational efficiency, the IntentCloud architecture decouples computational intelligence from storage management: language model inference and embedding generation are executed on a GPU- or CPU-capable workstation, while persistent storage and long-term file availability are maintained on a dedicated, always-on edge node. Secure remote access is provided through an encrypted tunneling mechanism, enabling users to reach their files from any location without exposing the local network or depending on third-party cloud storage providers.

The remainder of this paper is organized as follows. Section II presents a detailed literature survey covering semantic retrieval, Retrieval-Augmented Generation, intent classification, and edge-based deployment, organized thematically, and concludes with an explicit statement of the research gap and the motivation underlying the proposed IntentCloud system.

II. Detailed Literature Survey

Information retrieval research has progressively shifted from purely lexical matching techniques, such as BM25 and TF-IDF, toward dense vector representations produced by transformer encoders [2]. More recently, this trend has extended toward Retrieval-Augmented Generation (RAG) pipelines that couple a retriever with a generative language model to produce contextually grounded responses. This section reviews nineteen works relevant to the design of IntentCloud, organized into four thematic categories: semantic retrieval systems, Retrieval-Augmented Generation, intent classification, and edge AI with privacy-preserving deployment. Each subsection discusses the methodology, principal contributions, reported findings, and limitations of the reviewed works. A dedicated discussion of the research gap and the motivation for the proposed system follows at the end of this section.

A. Semantic Retrieval Systems

Vaswani *et al.* [2] introduced the transformer architecture and the self-attention mechanism, which subsequently became the foundational building block for nearly all modern sentence- and document-embedding models. Although this work does not itself propose a retrieval or indexing mechanism, its attention-based encoder underlies the sentence-embedding models used throughout the semantic retrieval literature, including in the present work.

Building on transformer-based representations, V. N. R. K *et al.* [3] proposed a semantic information retrieval pipeline that combines transformer embeddings with HDBSCAN clustering and generative re-ranking using large models such as GPT-3 and XLNet. Their methodology groups semantically similar documents prior to retrieval in order to reduce the search space, and their reported findings indicate improved topical coherence over purely lexical baselines. However, the clustering step introduces a reported 35% traversal penalty during index lookup, and the generative re-ranking stage depends on large, computationally expensive models that are unsuitable for real-time or resource-constrained deployment. The system additionally does not provide human-interpretable explanations for why a given result was retrieved.

Brundha and Meera [4] examined a vector-model-based information retrieval approach that transforms word embeddings to improve representational quality prior to similarity scoring. Their evaluation demonstrates that embedding transformation can meaningfully improve retrieval precision relative to raw embedding comparison. However, the approach entails high training cost for large embedding models and does not specify an upload pipeline, persistence layer, or deployment mechanism, limiting its applicability as a complete retrieval system.

Douze *et al.* [5] presented FAISS, a widely adopted library for efficient similarity search and clustering of dense vectors. FAISS supports multiple indexing structures, including flat, inverted-file, and product-quantization indexes, allowing developers to trade off search accuracy, memory footprint, and query latency according to application requirements. Its widespread adoption in the retrieval literature stems from its ability to perform approximate nearest-neighbor search over millions of vectors with sub-second latency on commodity hardware. As an algorithmic library, however, FAISS provides no persistence layer, concurrency control, or upload/serving infrastructure, and must be integrated into a larger system to function as a complete retrieval service.

Karthikeyan *et al.* [1] proposed a smart document search engine that automatically extracts metadata from uploaded files to support intelligent file discovery. Their methodology relies on a Kubernetes-based microservice architecture to orchestrate metadata extraction and indexing across distributed nodes. While this design supports horizontal scalability, it requires substantial infrastructure overhead and does not support conversational or iterative query refinement, limiting its usability for individual users operating on modest hardware.

B. Retrieval-Augmented Generation

Krishnan and Subramanian [6] implemented a document management system that performs semantic search using a Retrieval-Augmented Generation pipeline built around the commercial ChatGPT API. Their approach demonstrates that RAG can substantially improve the relevance of document search results compared to keyword matching, but it requires transmitting document content to a third-party cloud service, raising privacy concerns, and incurs recurring API costs that scale with usage.

Jacob *et al.* [7] constructed a conversational PDF-querying system using LangChain in combination with the OpenAI API. Their methodology segments documents into fixed-size chunks, typically on the order of 1500 tokens, before embedding and retrieval. While this chunking strategy is computationally efficient, it fragments the global context of a document, degrading performance on queries that require reasoning across sections that are separated by the chunk boundaries, in addition to depending on a paid, externally hosted model.

Sawarkar *et al.* [8] proposed Blended RAG, which combines dense vector retrieval with sparse, keyword-based retrieval and hybrid query formulations. Their experiments show that blending retrieval strategies improves accuracy relative to either method used in isolation, particularly on datasets with rich metadata. However, the dense index component is reported to require approximately 50 GB of memory for a five-million-document corpus, and the hybrid retrieval strategy degrades in performance on metadata-poor datasets.

Rafi *et al.* [9] developed a document search chatbot using RAG that depends on a commercial large language model API. Although the system produces coherent conversational answers grounded in retrieved passages, it returns short synthesized responses rather than the original source file, making it unsuitable for use cases in which the underlying document itself, rather than a summary of its contents, is the desired output.

Tiwari *et al.* [10] fine-tuned T5 and BERT models for question answering and document retrieval tasks, demonstrating that task-specific fine-tuning can improve retrieval and answer-generation accuracy over zero-shot use of pretrained models. This approach, however, requires access to GPU or TPU infrastructure for both training and serving, which constrains its deployment to organizations with substantial computational resources.

Saha *et al.* [11] introduced QuIM-RAG, which improves retrieval by generating candidate questions from document passages offline and matching user queries against this inverted question index. Their results show improved question-answering accuracy relative to standard passage-retrieval RAG. The offline question-generation step, however, depends on a commercial LLM, and the resulting index is static, requiring manual regeneration whenever new documents are added.

Cejas *et al.* [12] conducted an empirical study tracing the impact of embedding quality on downstream RAG performance, finding that larger embedding models improve retrieval accuracy but at the cost of increased GPU memory footprint and inference latency; the study also finds that restricting retrieval to a single top result (top-k=1) reduces the contextual richness available to the generative stage.

C. Intent Classification

Sriram *et al.* [13] proposed a method for leveraging local large language models to perform secure, in-system task automation through prompt-based classification of user commands. Their approach demonstrates that locally hosted models can perform command classification without transmitting data externally, preserving privacy. However, their evaluation reports a 14% task misclassification rate, and the system is confined to a single local machine with a command-line interface only, without a graphical interface or remote accessibility.

Kuchlous and Kadaba [14] developed a short-text intent classification method for conversational agents operating within a therapy chatbot, restricting classification to four fixed discourse categories. Their results indicate strong performance within this narrow, closed-set classification task, but the method does not generalize to open-ended intent parsing and includes no mechanism for file or document retrieval.

Puspitasari *et al.* [15] proposed an intent-aware RAG framework that combines intent classification with semantics-preserving chunking to improve retrieval relevance. Their methodology assumes that relevant information is typically localized within a single chunk or a small neighborhood of chunks, an assumption that does not hold for multi-hop queries requiring reasoning across content that spans multiple, non-adjacent sections of a document; the framework additionally does not address file delivery.

D. Edge AI, Hardware Acceleration, and Privacy-Preserving Deployment

Nan *et al.* [16] proposed an intelligent file query system integrating natural language understanding with information retrieval (DSMR). Their architecture achieves strong retrieval accuracy by combining a large-scale NLU model with a dedicated IR component, but the reported hardware configuration requires dual NVIDIA A100 GPUs and 128 GB of system memory, placing the approach well beyond the reach of individual users or small organizations operating on consumer-grade hardware.

Holubiev *et al.* [17] examined the use of autonomous agents, employing ReAct, Reflexion, and Chain-of-Thought reasoning loops, for knowledge management in contact-center environments. While the multi-agent design improves the quality of organized textual knowledge, the recursive reasoning loops introduce substantial latency, and the work does not include quantitative information-retrieval benchmarking, making it difficult to assess retrieval effectiveness relative to simpler architectures.

Nickolls *et al.* [18] described the CUDA parallel programming model, which enables general-purpose computation on NVIDIA GPUs and underlies GPU-accelerated inference in the majority of modern deep learning systems. Apple Inc. [19] similarly documented the Metal Performance Shaders (MPS) framework, which provides comparable GPU acceleration on Apple Silicon hardware. Neither work is tied to a specific document-retrieval or LLM-serving workflow; rather, each provides a hardware acceleration backend that must be integrated into a larger inference pipeline to be useful for retrieval tasks.

E. Research Gap

The literature reviewed above reveals several unresolved and recurring limitations that collectively motivate the present work:

- **Dependence on commercial cloud APIs.** A substantial portion of RAG-based systems [6], [7], [9], [11] require transmitting document content to externally hosted, commercial large language models, exposing potentially sensitive data to third parties and incurring recurring usage costs.
- **Limited privacy preservation.** Systems that rely on cloud-hosted inference cannot guarantee that document content remains within the user's control, which is particularly problematic for personal or confidential documents.
- **Lack of intent-aware retrieval for personal file management.** Existing intent classification approaches [13], [14], [15] are designed for conversational agents or narrow, closed-set discourse categories rather than for open-ended, personal document retrieval.
- **Inability to return the original file.** Several RAG-based systems [6], [9], [11] return synthesized textual answers rather than the source document, limiting their usefulness in contexts where the file itself, rather than a summary, is required.
- **Reliance on resource-intensive hardware.** High-accuracy systems such as DSMR [16] and Blended RAG [8] depend on GPU-cluster-scale infrastructure or

TABLE I
SUMMARY OF RESEARCH GAPS IDENTIFIED IN THE LITERATURE

Research Gap Category	Description	Representative Works
Dependence on commercial cloud APIs	Retrieval or generation pipeline requires an externally hosted, paid large language model, exposing document content to third parties	[6], [7], [9], [11]
Limited privacy preservation	Document content or embeddings leave the user's local environment during processing	[6], [7], [9]
Lack of open-ended, personal-file intent understanding	Intent classification restricted to narrow, closed-set categories or conversational-agent contexts	[13], [14], [15]
Inability to return the original file	System produces a synthesized textual answer rather than delivering the source document	[6], [9], [11]
Reliance on resource-intensive hardware	Accurate retrieval depends on GPU-cluster-scale infrastructure or memory-heavy indexes	[8], [16]
Absence of complete, self-hosted deployment	Algorithmic contribution lacks an accompanying upload, indexing, and serving layer	[2], [5]
Limited cross-platform inference support	System assumes a fixed hardware configuration rather than portable acceleration	[18], [19]
Limited secure remote accessibility	Locally hosted system is confined to a single machine without remote access	[13]

memory-heavy indexes, restricting their accessibility to organizations with substantial computational resources.

- **Absence of lightweight, self-hosted deployment.** Foundational tools such as FAISS [5] and transformer embeddings [2] provide the algorithmic building blocks for retrieval but are not packaged as complete, self-hosted systems with upload, indexing, and serving capability.
- **Limited support for cross-platform inference.** Most reviewed systems assume a fixed hardware configuration and do not address portability across NVIDIA, Apple Silicon, or CPU-only environments.
- **Limited support for secure remote accessibility.** Privacy-preserving, locally hosted systems [13] are typically confined to a single machine, without a mechanism for secure remote access from other locations or devices.

F. Research Motivation

These limitations collectively motivate the development of IntentCloud, which integrates intent understanding, semantic search, local model inference, edge-based persistent storage, and secure remote accessibility into a single, unified architecture. Rather than treating retrieval accuracy, privacy, hardware accessibility, and remote availability as separate concerns, as is common throughout the reviewed literature, IntentCloud is designed to address these requirements concurrently within a self-hosted deployment suitable for individual users and small teams operating on consumer-grade hardware.

Table I summarizes the principal categories of research gaps identified in the literature and indicates the representative works in which each gap is most prominent.

Therefore, the reviewed literature demonstrates that although significant advances have been made in semantic retrieval and Retrieval-Augmented Generation, no existing solution simultaneously addresses privacy preservation, intent-aware retrieval, lightweight deployment, and secure remote accessibility. This gap forms the basis of the proposed IntentCloud framework.

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